

Unit 7, Lesson 1: Factoring and the Distributive Property



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Targets

- ☐ I can rewrite a polynomial using the distributive property by factoring out a common factor.
- ☐ I can determine the greatest common monomial factor.



7.1.1 Warm-Up: Electric Motorcycle Engineer

1. Marc Fenigstein says, “Open-ended problems are the problems students will face in the real world ... Forcing students to learn how to function in ambiguity and find their own answers ... and that there isn’t necessarily any one right answer, but there might be better or worse answers ... I think that is really valuable. Don’t be afraid to find your own path ... You’re not going to get it right on the first try, but you can make almost any job your dream job by looking for opportunities to do more, to go beyond ...”
 - a. What stands out to you the most in this statement?
 - b. Why does that part of the statement stand out to you?



7.1.2 Exploration: The Distributive Property in Factoring

1. Multiply $9(10 + 6)$ using the distributive property.
2. Discuss with your partner how $9(10 + 6)$ shows that 9 and 16 are factors of 144.
3. Discuss with your partner how multiplying $8(287)$ can be done more easily using the distributive property. Explain or show your work.
4. Rewrite the expression $-7(2x - 5)$ using the distributive property.

5. Complete the table.

Expression	Factors	Product
$9(10 + 6)$	9 and 16	144
$-7(2x - 5)$	-7 and $2x - 5$	
$(8 - 4x)10$		
$-12(7 - 3x)$		
$2(6x^2 + 8x)$		
$4x(3x + 4)$		
$(6x + 8) \cdot 2x$		

6. What do you notice about the product of each of the expressions $2(6x^2 + 8x)$, $4x(3x + 4)$, and $(6x + 8) \cdot 2x$?

Expressions are **equivalent** if they have the same value no matter what number is substituted in for each of the variables.

7. The expression $14x^2 + 28x^3$ is the product of each of the following expressions.

$$2(7x^2 + 14x^3)$$

$$x^2(14 + 28x)$$

$$7x^2(2 + 4x)$$

List the factors for each expression.

Factors of $2(7x^2 + 14x^3)$		Factors of $x^2(14 + 28x)$		Factors of $7x^2(2 + 4x)$	

8. Write two more expressions that are equivalent to $14x^2 + 28x^3$.
9. Compare your expressions with your partner's expression. Write the expressions that you and your partner do not have in common.

10. Complete the table for the factors of $14x^2 + 28x^3$ using two other expressions you created.

Monomial Factor	Binomial Factor
2	$(7x^2 + 14x^3)$
x^2	$(14 + 28x)$
$7x^2$	$(2 + 4x)$



7.1.3 Guided Instruction: Common Monomial Factors

The table in the exploration shows there are several different ways to rewrite the expression $14x^2 + 28x^3$.

1. What monomial factor is the greatest common monomial factor of $14x^2 + 28x^3$?
2. Discuss with your partner how to determine if a monomial factor is the greatest common monomial factor of a given polynomial. Summarize your discussion.
3. Determine the greatest common factor for 24, -36, and -54.
4. Determine the greatest common factor for x^7 , x^{10} , and x^{13} .

5. Determine the greatest common factor for x^3y^3 , x^2y^8 , and xy^4 .
6. Work with your partner to complete the table. If an expression does not have one of the greatest common factors listed, write N/A.

Expression	Greatest Common Numeric Factor	Greatest Common Variable Factor	Greatest Common Monomial Factor
$25w^3 - 40w^2 + 60w$			
$81g + 108g^2 - 12g^4$			
$7m^3n^2 + 56m^2n^3 - 21mn$			
$12p^2q + 16q^2 - 64p$			



7.1.4 Guided Practice: Factoring With the Distributive Property

- Which of the following expressions are equivalent to $8a^3 - 98a$? Select all that apply.
 - ☐ $a(8a^2 - 98)$
 - ☐ $2(4a^3 - 49a)$
 - ☐ $2a(4a^2 - 49)$
 - ☐ $2a(4a^3 - 49a)$
 - ☐ $4a(2a^2 - 49a)$
 - ☐ $8a^3(2 - 98a)$
- Write three equivalent expressions for $18x^3 - 9x^2 - 15x$. One of the expressions should include the greatest common monomial factor.
- Write an equivalent expression for $2c^3d^3 + 14c^2d + 18cd$ that uses the greatest common monomial factor.



7.1.5 Your Turn!

- Write three equivalent expressions for $72b^2k^3 - 64b^5k^2 + 32b^8k^6$. One of the expressions should include the greatest common monomial factor.
- Which expressions are equivalent to $48r^3 + 18r^2$? Select all that apply.
 - ☐ $3(16r^3 + 6r^2)$
 - ☐ $6r(8r^2 + 3r)$
 - ☐ $2r^2(24r + 9)$
 - ☐ $4(12r^3 + 4r^2)$
 - ☐ $6r^2(8r + 3)$



7.1.6 Wrap-Up: Ticket Out the Door

1. Write three equivalent expressions for $12a^3b^5 - 16a^2b^2 + 20a^3b$. One of the expressions should include the greatest common monomial factor.

Unit 7, Lesson 2: Factoring a Trinomial With Algebra Tiles



Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.



Learning Target

- ☐ I can use algebra tiles to factor a trinomial with a leading coefficient of 1.